Python BMI Calculator: A Beginner's Guide

A step-by-step introduction to writing real Python code — starting with a simple, practical health calculation.

What is BMI?

Definition

BMI (Body Mass Index) is a measure of body fat based on a person's height and weight.

Formula

BMI = weight (kg) ÷ height $(m)^2$

Purpose

Used to screen for weight categories: underweight, normal, overweight, and obese.

The Basic Formula in Python

Start with the simplest approach: two inputs, one calculation.

- **Variables needed:** `weight` and `height`
- **Operation:** Divide weight by height squared
- **Output:** BMI value rounded to 2 decimal places

Even the most complex programs start with simple variables and math!

Simple Function Example

```
def calculate_bmi(weight, height):
    bmi = weight / (height ** 2)
    return round(bmi, 2)
```

How it works

- Takes `weight` (kg) and `height` (m) as parameters
- Computes the BMI using Python's `**` exponent operator
- Returns BMI rounded to 2 decimal places using `round()`

Using the Function

```python
weight = 70     # kilograms
height = 1.75   # meters

result = calculate_bmi(weight, height)
print(f"Your BMI is: {result}")
```

What's happening here?

- Assign real values to `weight` and `height`
- Call the function and store the result
- Print the result to the user with an f-string

Running this code prints: `Your BMI is: 22.86`

Adding Categories

BMI alone is just a number — categories give it meaning. Here are the standard ranges:

Underweight

BMI < 18.5

Normal Weight

18.5 ≤ BMI < 25

Overweight

25 ≤ BMI < 30

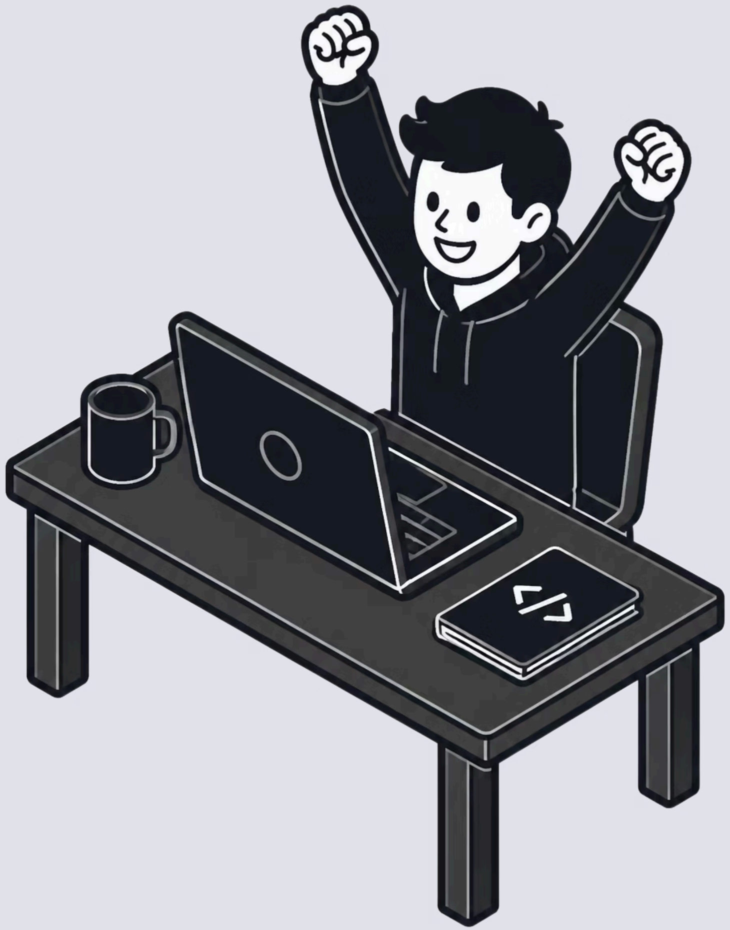
Obese

BMI ≥ 30

Function with Categories

```python
def bmi_category(bmi):
    if bmi < 18.5:
        return "Underweight"
    elif bmi < 25:
        return "Normal"
    elif bmi < 30:
        return "Overweight"
    else:
        return "Obese"
```

Key concepts used



Complete Program

Put it all together — a fully working BMI calculator in just a few lines of Python.

01

Get User Input

Use input() and float() to collect weight and height from the user.

02

Calculate BMI

Call calculate_bmi(weight, height) and store the result.

03

Display Category

Pass the result into bmi_category() and print both the number and label.

04

What's Next?

Expand with error handling, unit conversion, or even a graphical interface!